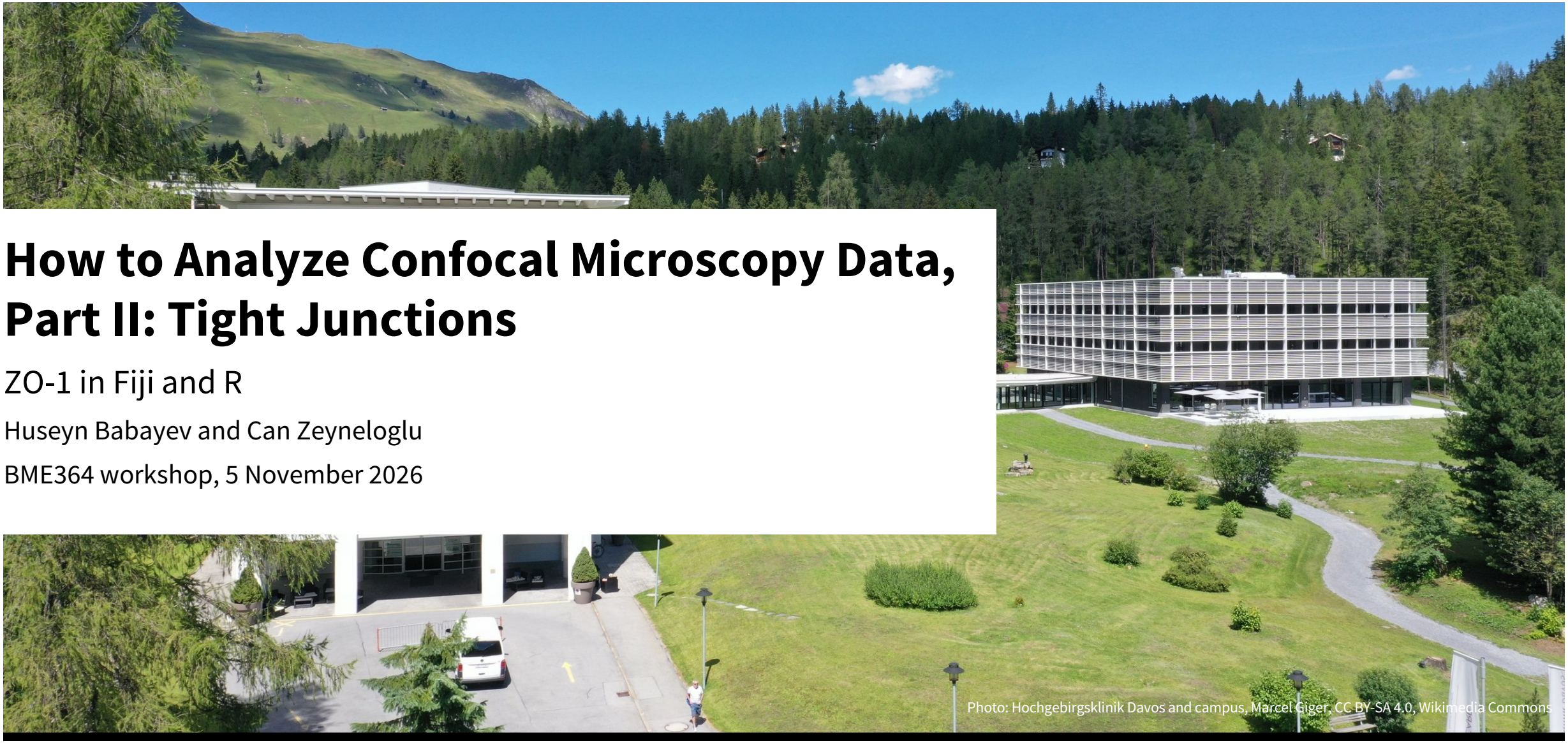
Universität Zürich[UZH]

Swiss Institute of Allergy and Asthma Research (SIAF)



# How to Analyze Confocal Microscopy Data, Part II: Tight Junctions

## ZO-1 in Fiji and R

Huseyn Babayev and Can Zeyneloglu

BME364 workshop, 5 November 2026

Photo: Hochgebirgsklinik Davos and campus, Marcel Giger, CC BY-SA 4.0, Wikimedia Commons

# Teaching plan

## 4 Nov: Part I

**Fiji basics with JC-1**

- Pixels, bit depth, integrity
- Threshold → watershed → Analyze Particles
- Red : green ratio per cell

## Today: Part II

**Tight junctions with ZO-1**

- Z-stack → projection
- ZO-1 and DAPI mean per field
- Fields → wells → statistics in R

## Davos: Group I

**Your own experiment**

- Caco-2: P80 0, 0.02, 0.1, 0.5 %
- ZO-1 + DAPI, 96-well glass-bottom plate
- 5 wells × 3 random fields, confocal

**Course materials**

github.com/drbabayev/UZH-BME364-G1-share

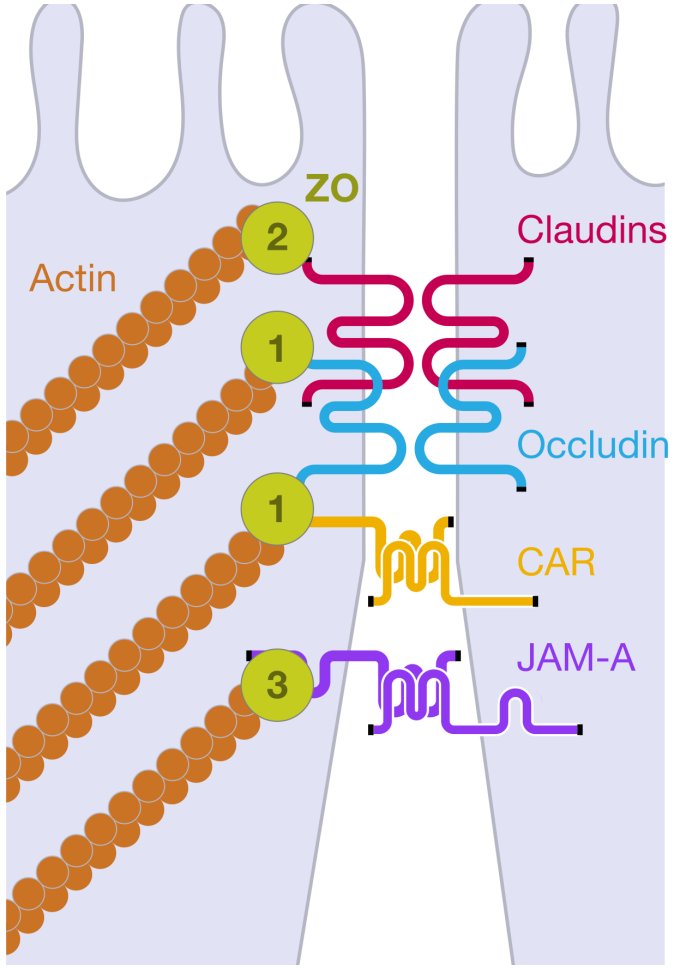
# Measuring ZO-1 in Fiji

Theory, data structure, algorithm, implementation

01

# ZO-1 is the scaffold of every tight junction

Theory | Data | Algorithm | Implementation




Zeyneloglu, Babayev et al., FEBS Lett 2025, Fig. 2B;
doi:10.1002/1873-3468.70113

## ZO-1 is the scaffold

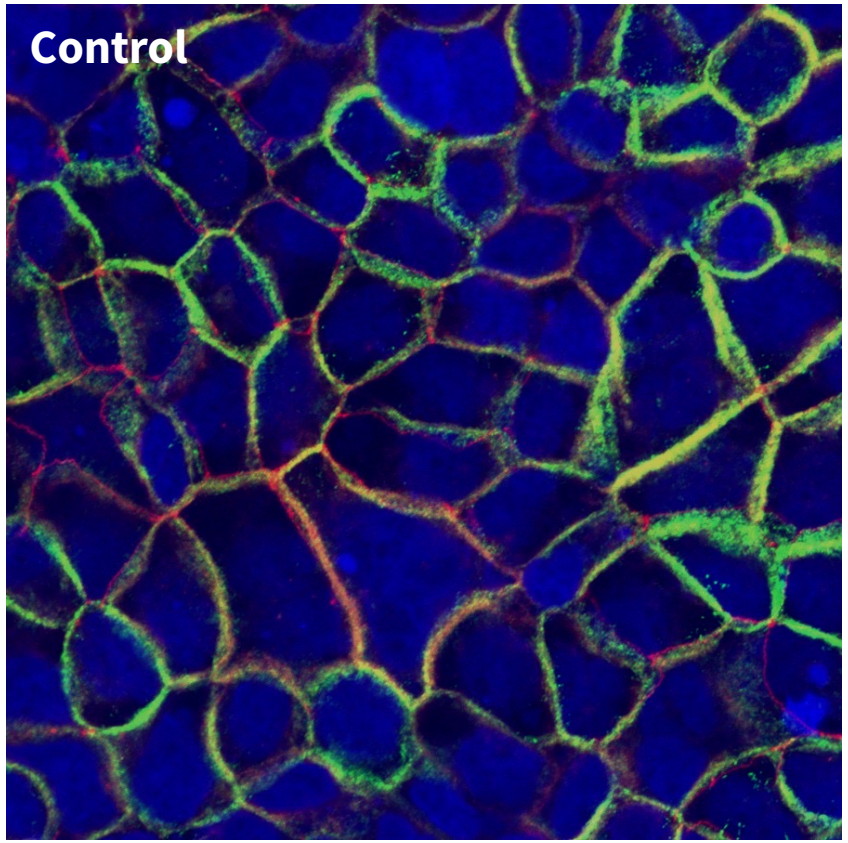
ZO-1, ZO-2 and ZO-3 bind claudins, occludin, CAR and JAM-A and link them to actin: ZO-1 outlines every cell as a continuous honeycomb.

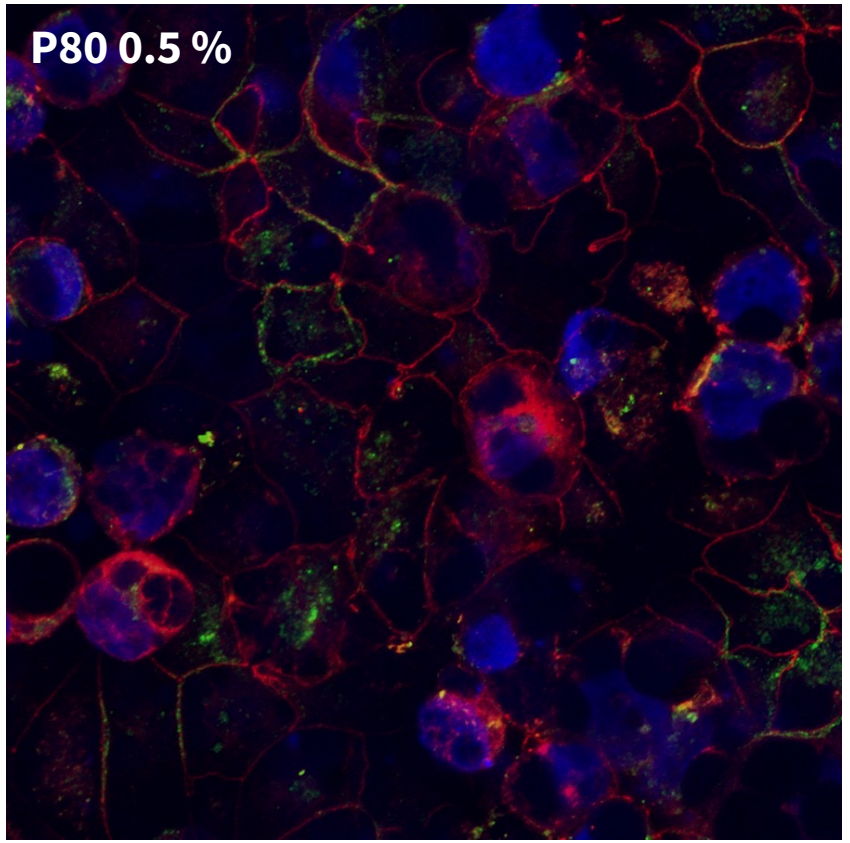
## P80 opens the barrier

Polysorbate 80 is a detergent: the junctions loosen, ZO-1 falls and the honeycomb breaks up. At high doses the cells die.

# What P80 does to a real monolayer






Real fields, earlier experiment: Caco-2, occludin + claudin-1, DAPI blue; max projection

**Control**

Closed honeycomb: every cell outlined, junctions continuous.

**P80 0.5 %**

Network broken, weaker junction signal, cells round up: MTT shows no living cells.

# Immunofluorescence: from ZO-1 to a signal



**1 Fix**
4 % PFA, 15 min

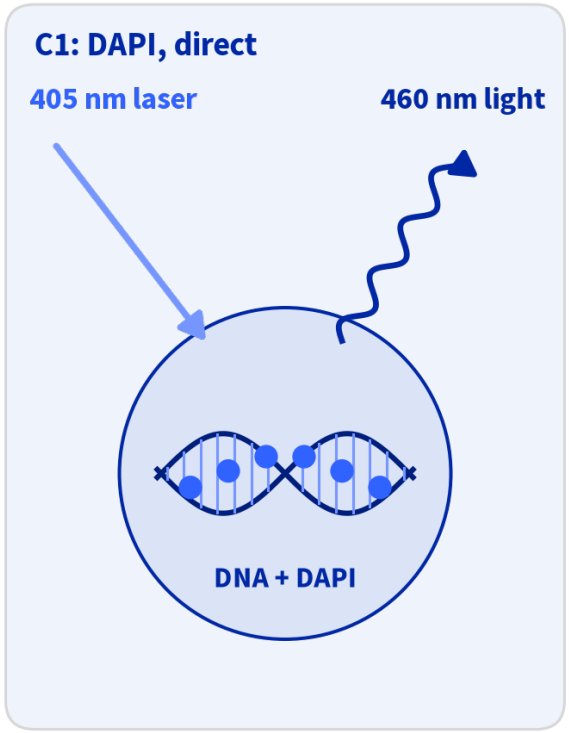
**2 Permeabilise**
0.1 % Triton X-100, 10 min

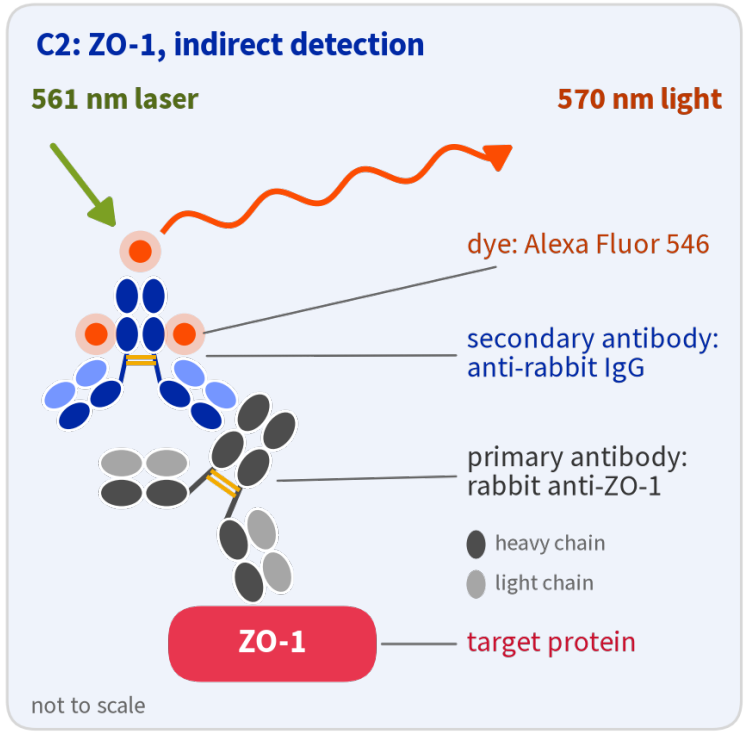
**3 Block**
3 % BSA, 1 h

**4 Primary**
rabbit anti-ZO-1, 1:100

**5 Secondary**
anti-rabbit Alexa 546, 1:500

**6 DAPI**
1 µg/mL, nuclei




## Indirect detection

Several labelled secondaries bind each primary: a brighter signal.

## Light in, light out

One laser per dye, longer light out: C1 DAPI (405 nm), C2 ZO-1 (561 nm).

## Confocal pinhole

Blocks out-of-focus light: thin optical sections; a z-stack covers the junctions.

## Same settings

Same laser, gain, pinhole and zoom for every well; no saturated pixels (HiLo LUT).

## No-primary background

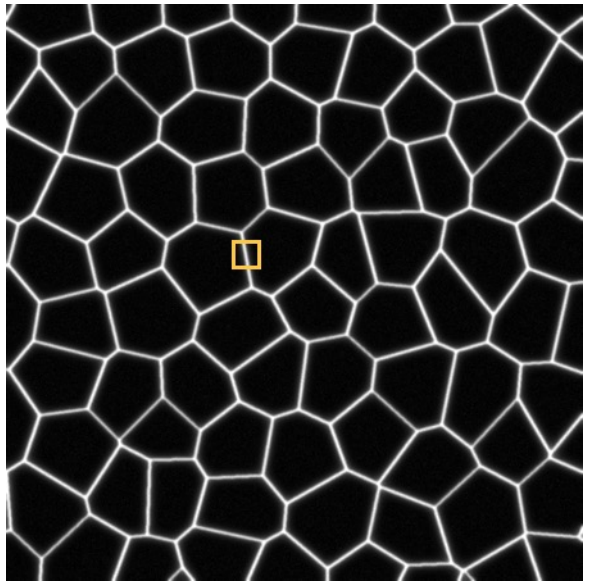
Secondary only, same settings: its mean is subtracted from every field.

## Two means per field

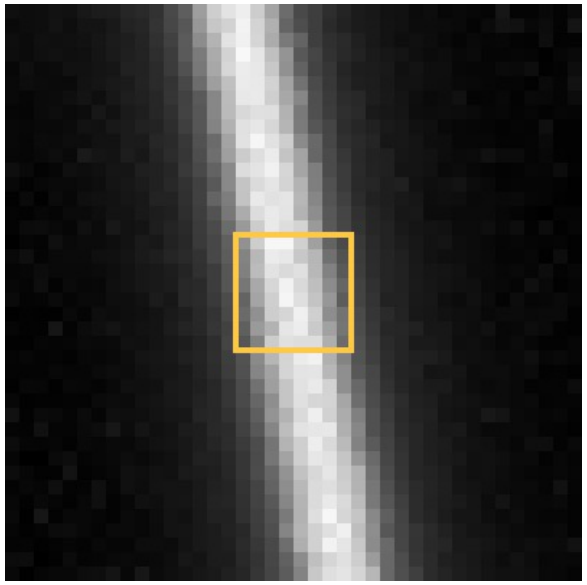
DAPI and ZO-1, whole field, raw. **Read-out:** the ZO-1/DAPI ratio.

# An image is a grid of numbers





ZO-1 channel, 900 × 900 px = 90 × 90 µm



40 × 40 pixels (4 × 4 µm): each square is one pixel

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1726 | 2352 | 2776 | 2774 | 2387 | 1881 | 1256 | 991 |
| 1581 | 2123 | 2666 | 2624 | 2294 | 2116 | 1391 | 1071 |
| 1555 | 2163 | 2576 | 2481 | 2541 | 2129 | 1565 | 1146 |
| 1454 | 1968 | 2448 | 2729 | 2472 | 2200 | 1670 | 1274 |
| 1252 | 1923 | 2366 | 2758 | 2535 | 2250 | 1778 | 1184 |
| 1327 | 1720 | 2356 | 2438 | 2709 | 2373 | 1690 | 1414 |
| 1206 | 1828 | 2068 | 2600 | 2673 | 2312 | 1898 | 1437 |
| 1118 | 1574 | 2340 | 2555 | 2523 | 2518 | 2035 | 1336 |

8 × 8 pixel values = what Fiji measures

### Junction

high values (≈ 2700) in the middle columns

### Cytoplasm

low values (≈ 1000) at the edges

### Fiji

the status bar shows the value under the cursor

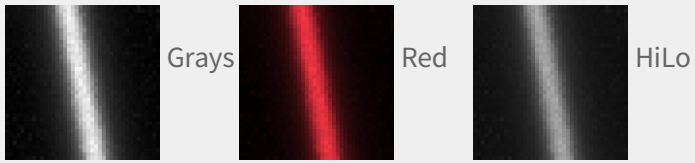
## Pixel size = calibration

0.1 µm per pixel, stored in the file metadata (Image › Properties). Area = pixel count × 0.01 µm².

## Bit depth = value range

8-bit 0–255, 12-bit 0–4095, 16-bit 0–65 535. A pixel at the maximum is saturated: its true value is lost.

## LUT = colors, not data

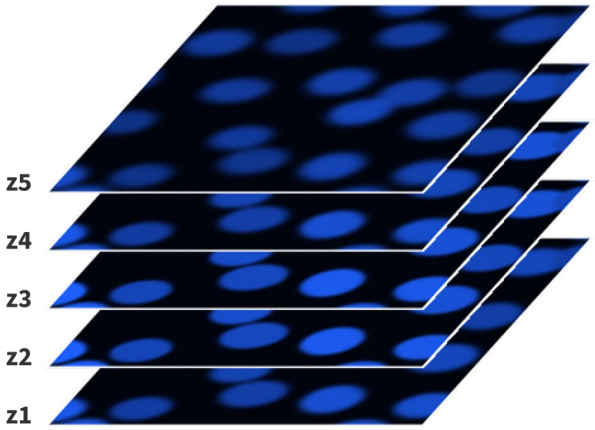



same numbers, HiLo: red = saturated, blue = 0

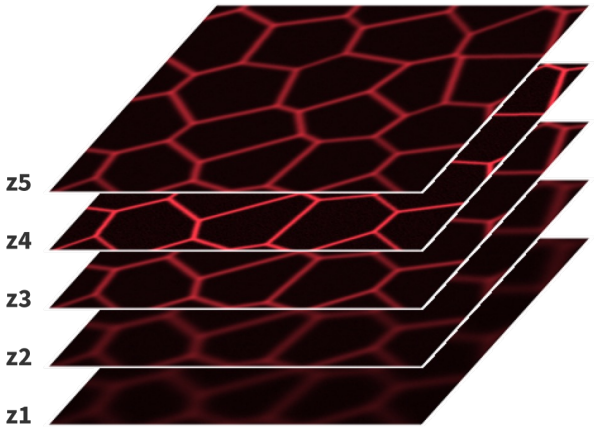
# A confocal file is a hyperstack: X, Y, C, Z



## C1: DAPI (nuclei)




## C2: ZO-1 (junctions)




5 z-slices, 0.5 µm apart, nuclei sharp at z2–z3, ZO-1 at z4 (apical)

## Image › Show Info: the file's dimensions

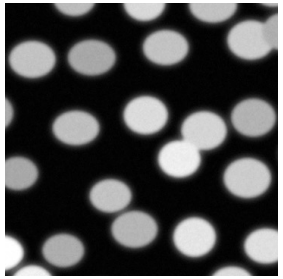
| Dimension | Size | Calibration |
|---|---|---|
| **X (width)** | 900 px | 0.1 µm/px → 90 µm |
| **Y (height)** | 900 px | 0.1 µm/px → 90 µm |
| **C (channels)** | 2 | DAPI, ZO-1 |
| **Z (slices)** | 5 | 0.5 µm steps |
| **T (frames)** | 1 | one time point |
| **Bit depth** | 16-bit | 2 bytes per value |

**900 × 900 × 2 C × 5 Z × 2 bytes = 16.2 MB**

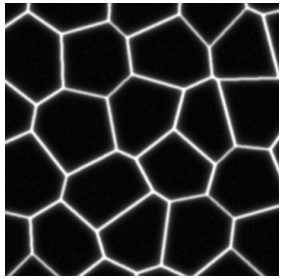
## Max projection: per pixel, the highest value along z

| z1 | z2 | z3 | z4 | z5 | → | MAX |
|---|---|---|---|---|---|---|
| 593 | 891 | 1582 | **2776** | 1576 | → | **2776** |

One ZO-1 pixel on a junction: in focus at z4. Use the same slices for every field.



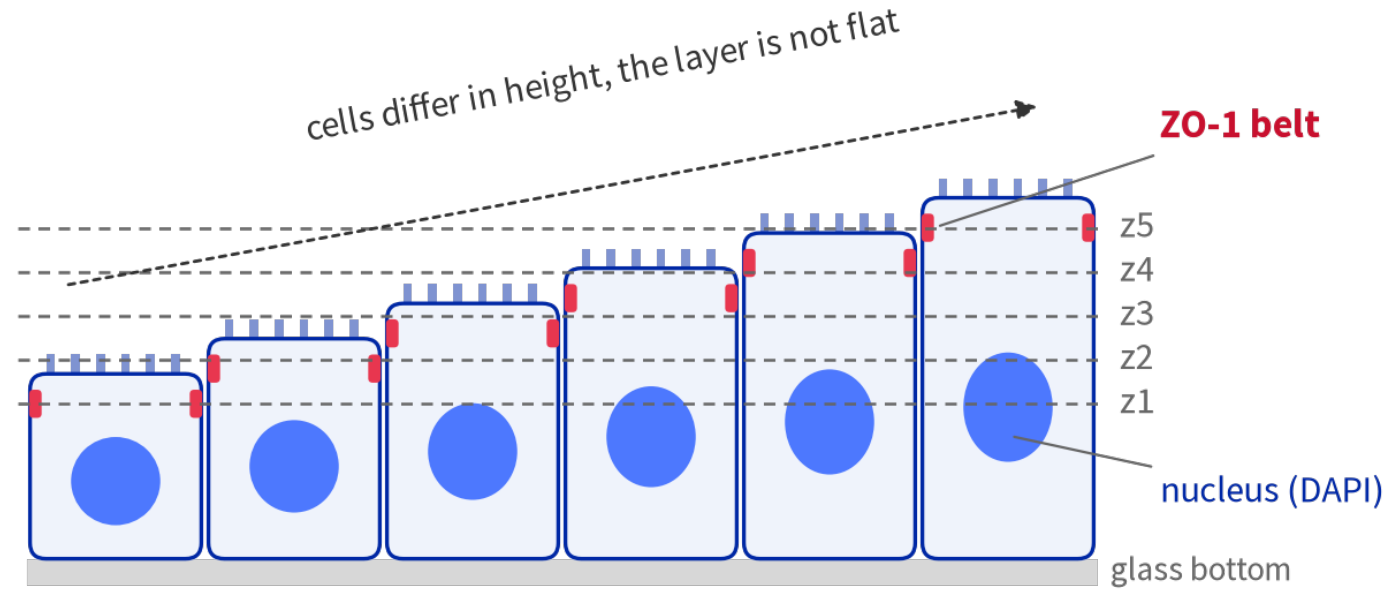
C1-MAX: DAPI

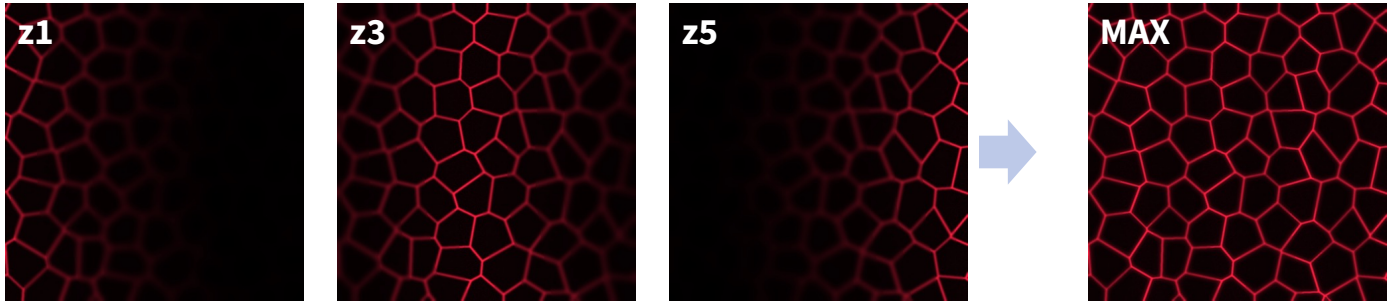


C2-MAX: ZO-1

# Why max projection: junctions sit at different heights






Simulated ZO-1, tilted control field: one slice is sharp in one band; MAX shows every junction

## Junctions at different heights

Cells differ in height and the layer is not flat: the ZO-1 belt lies at z1 in one cell and at z5 in another.

## One slice misses junctions

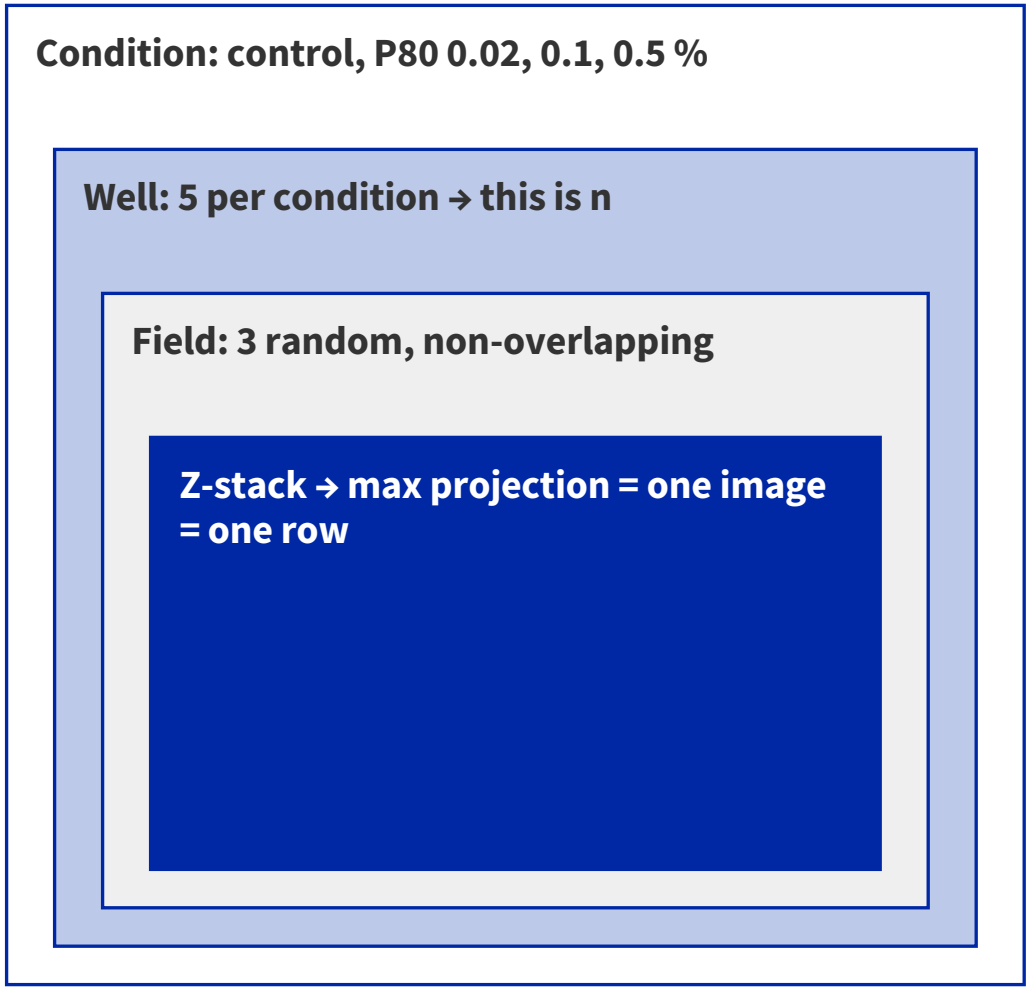
Out of focus, junctions blur and fade: an intact barrier would look broken.

## Max intensity projection

Per pixel, the brightest value along z: every junction at its best focus. Same slices for every field.

# From plate to Results table






**Fiji will give us results.csv: measurements, one row per image**

| Label | DAPI_Mean | ZO1_Mean |
|---|---|---|
| zo1_001.tif | 419.6 | 871.4 |
| zo1_031.tif | 442.8 | 871.9 |
| zo1_061.tif | 504.8 | 114.9 |

**You write image_metadata.csv: what each image is**

| file | plate_well | condition | p80_pct | well | field | primary_ab |
|---|---|---|---|---|---|---|
| zo1_001.tif | B2 | ctrl | 0 | well1 | 1 | ZO-1 |
| zo1_031.tif | D2 | P80-0.1 | 0.1 | well1 | 1 | ZO-1 |
| zo1_061.tif | B7 | noprimary | 0 | well1 | 1 | none |

- **Metadata table describes the data:** one row per image, filled in while imaging
- **File name = key only:** it links the two tables and carries no design
- **Tidy format:** one observation per row, one variable per column

# Per field: project, split, measure



### 1 Project

`Image › Stacks › Z Project… Max Intensity`

Same slices for every field

### 2 Split channels

`Image › Color › Split Channels`

C1 = DAPI, C2 = ZO-1

### 3 DAPI mean

`C1: Analyze › Measure`

Whole field, raw

### 4 ZO-1 mean

`C2: Analyze › Measure`

Whole field, raw

**One row per field → results.csv: Label, DAPI_Mean, ZO1_Mean**

In R: ratio = (ZO1_Mean − no-primary ZO-1) ÷ DAPI_Mean, then the mean of the 3 fields of each well

**Avoid:** measuring after auto-contrast or “Enhance Contrast”, and “Mean / Area × 1000”: Mean is already per pixel; dividing by area again just penalises large regions.

# ZO-1 amount: the ZO-1/DAPI ratio



$$\text{ZO-1/DAPI} = (\text{ZO-1 mean}_{\text{field}} - \text{ZO-1 mean}_{\text{no-primary}}) \,/\, \text{DAPI mean}_{\text{field}}$$

## We use: ZO-1/DAPI ratio

- ZO-1 per nuclear DNA ≈ ZO-1 per cell
- Gaps lower ZO-1 and DAPI together: they cancel
- Two whole-field means: no segmentation, nothing to tune

## Alternative

- Segment nuclei as in Part I: Otsu → watershed → Analyze Particles
- Grow them by ~8 µm = area covered by cells
- ZO-1 mean inside that area; nuclei count as a bonus

## Rationale

- No watershed or growth radius to check per image
- Fair while cells stay viable (MTT ≥ 94 % up to 0.1 %)
- DAPI raw and never saturated

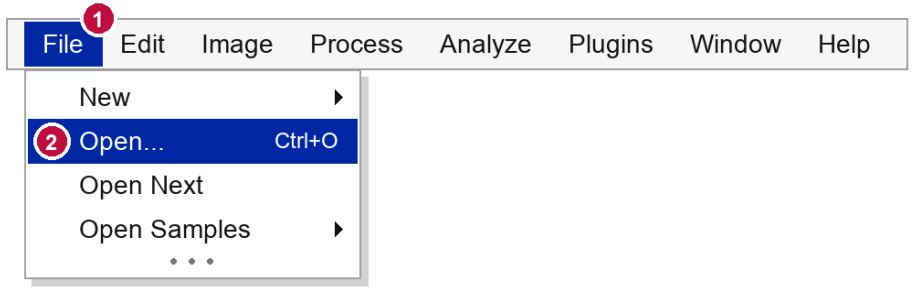
**Assumption:** every nucleus gives the same DAPI in all groups. Condensed, apoptotic nuclei are brighter: at 0.5 % P80, where the cells die, read the ratio with care.

# Fiji manually 1: open and project

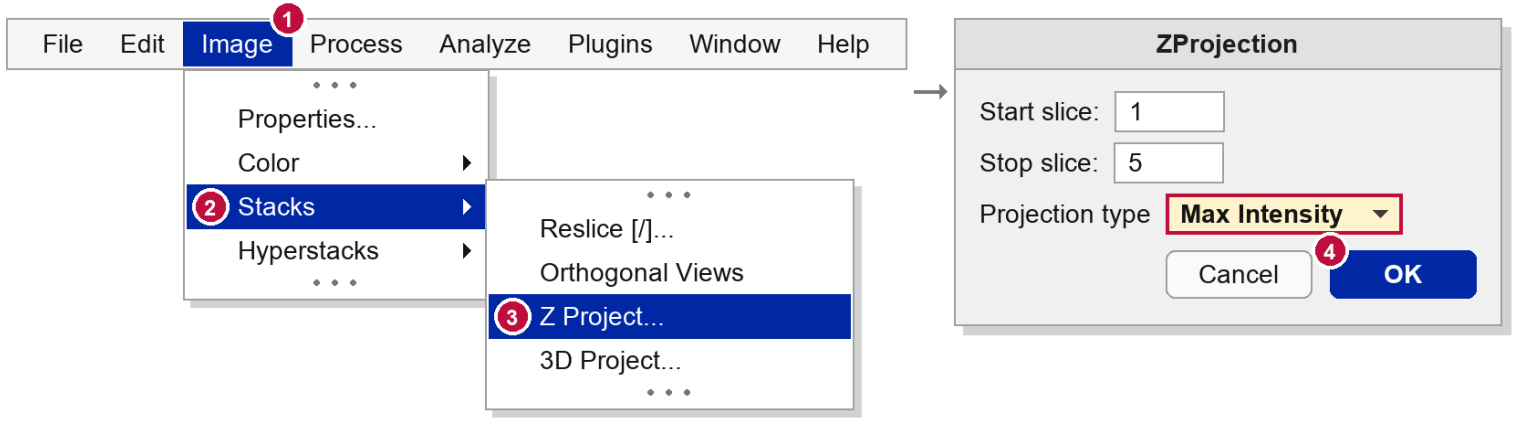


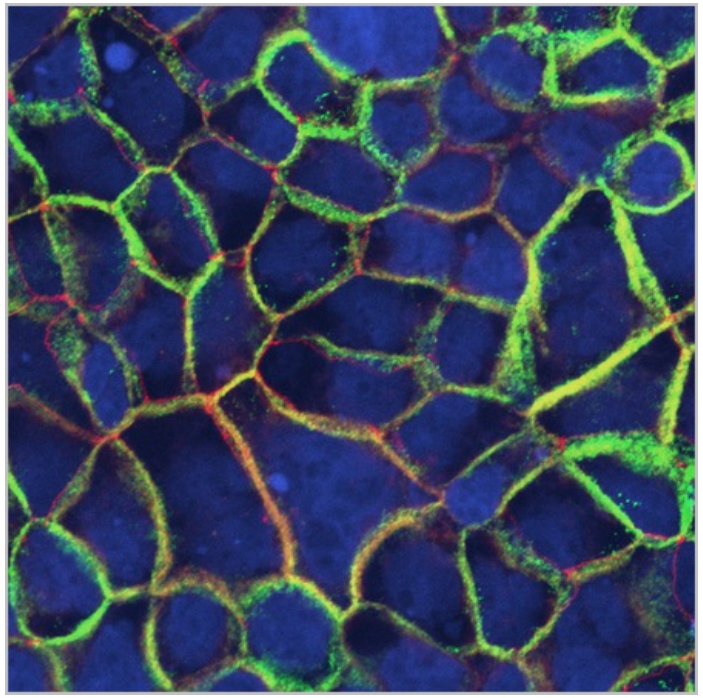
## 1 Open the TIFF, project the z-stack

**Open the .tif (2 channels × 5 slices), or drag it onto Fiji**




**Max Intensity over all 5 slices**



if_001.tif: real field, already projected

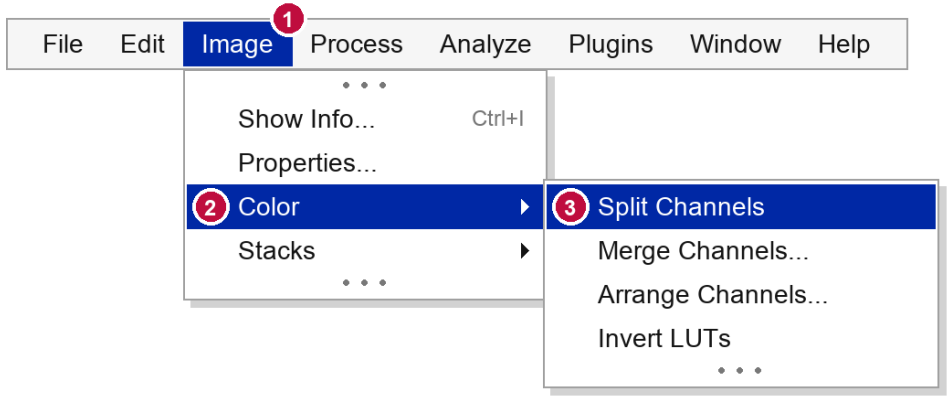
Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

# Fiji manually 2: split the channels

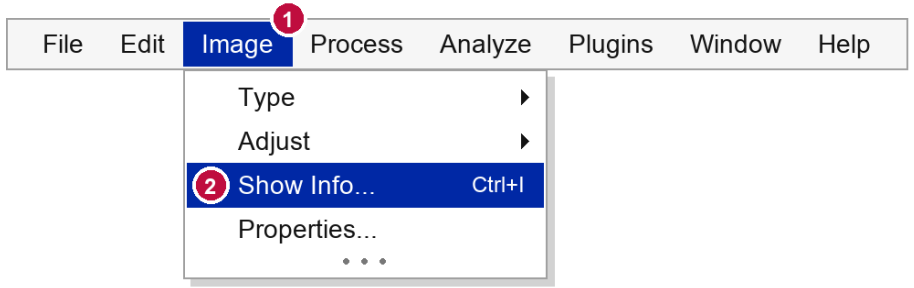


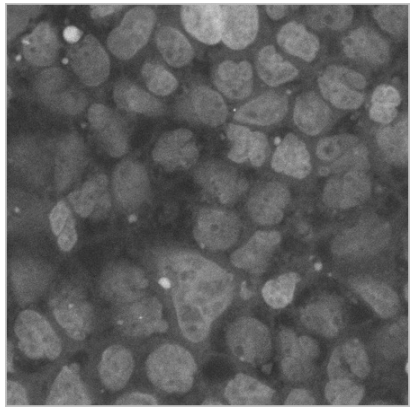
## 2 Split the channels, check the file

**One grayscale image per stain: C1 = DAPI, C2 = ZO-1**

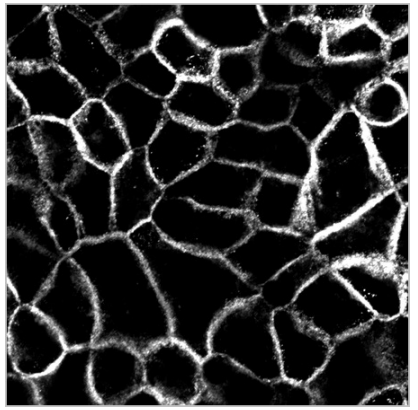



**Check: pixel size 0.1 µm, 2 channels, 5 slices**



C1: DAPI



C2: junction stain

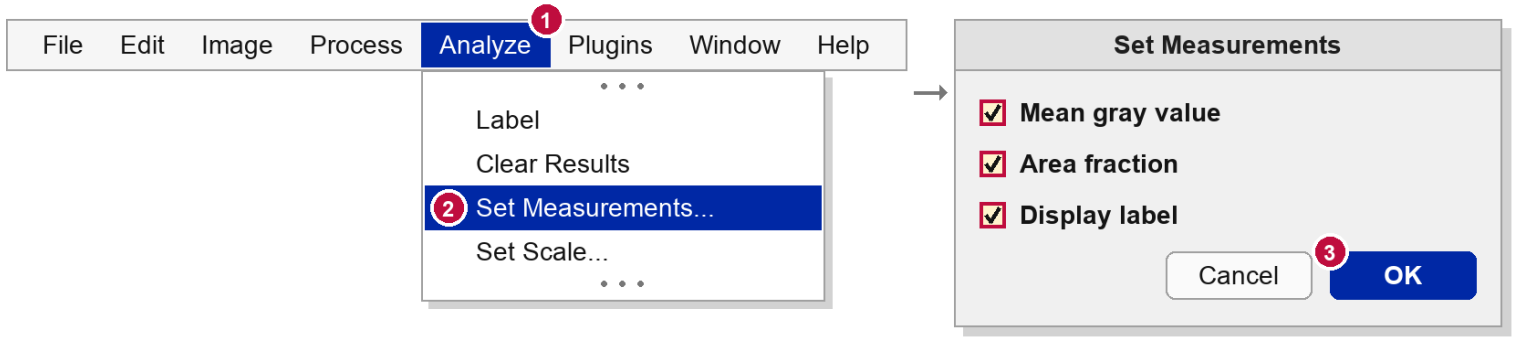
Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

# Fiji manually 3: DAPI for the ZO-1/DAPI ratio

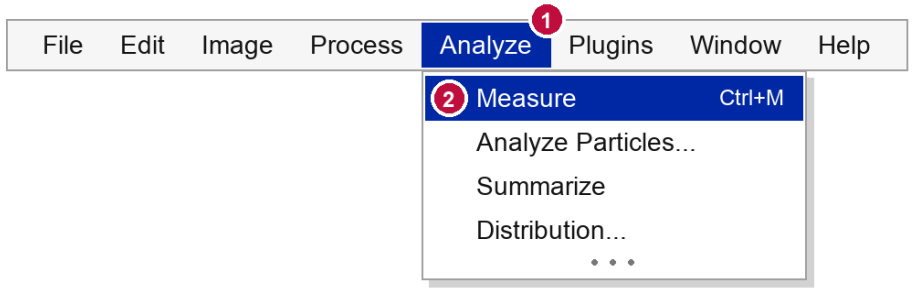


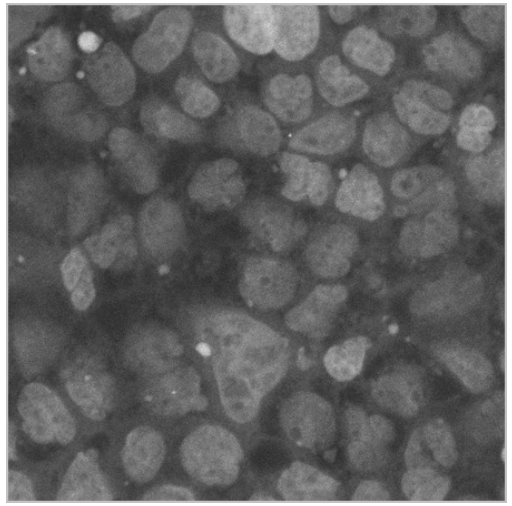
## 3 Measure DAPI on C1

**Once: Mean gray value, Area fraction, Display label**




**Click C1 (DAPI), nothing selected: Measure, raw values**



C1: raw

**91.8**

DAPI mean, whole field

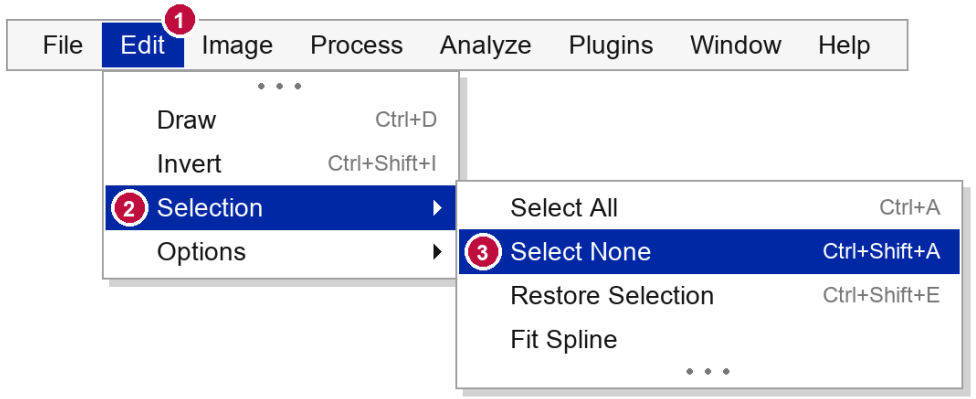
Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

# Fiji manually 4: ZO-1 for the ratio

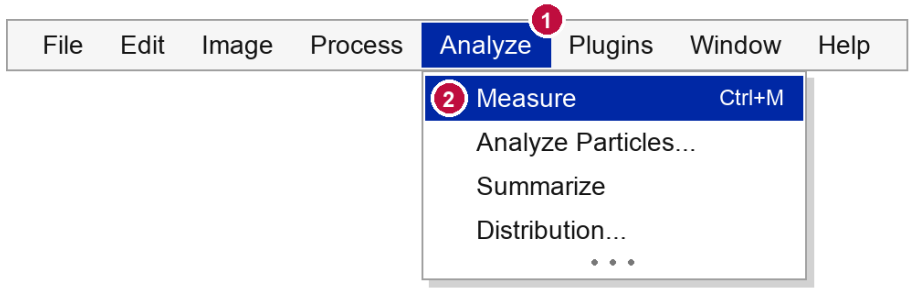


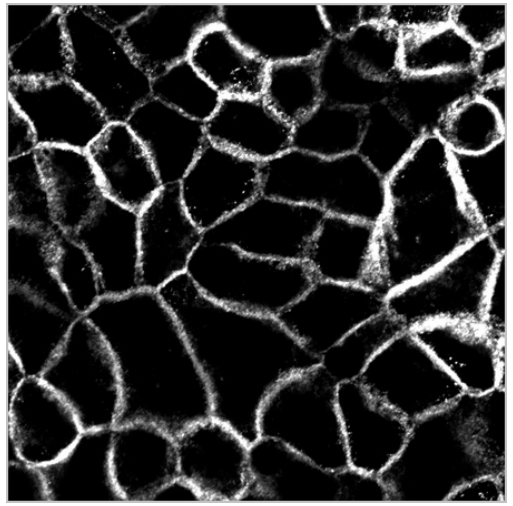
## 4 Measure raw ZO-1 on C2

### Click C2 (ZO-1): make sure nothing is selected




### Measure the whole field: raw values



C2: raw

**24.3**

C2 mean, whole field

**R:** (ZO-1 − no-primary) ÷ DAPI

Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

# Fiji manually 5: one row per field



## 5 Same steps on every field

**1–2** Project, split the channels

**3** C1 (DAPI): Measure, whole field, raw

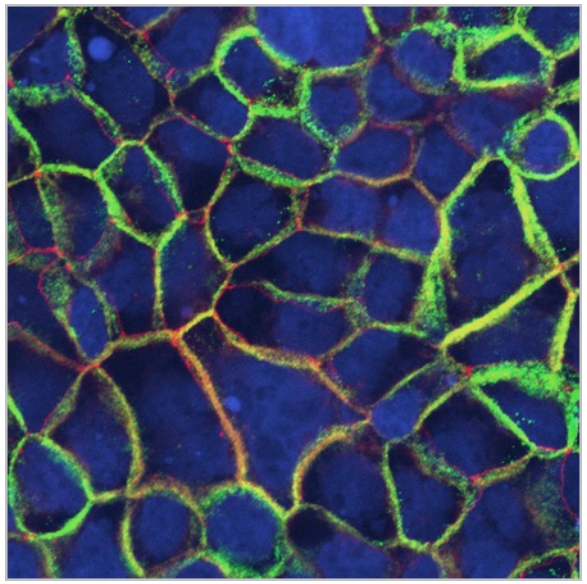
**4** C2 (ZO-1): Measure, whole field, raw

### Results table: one row per field

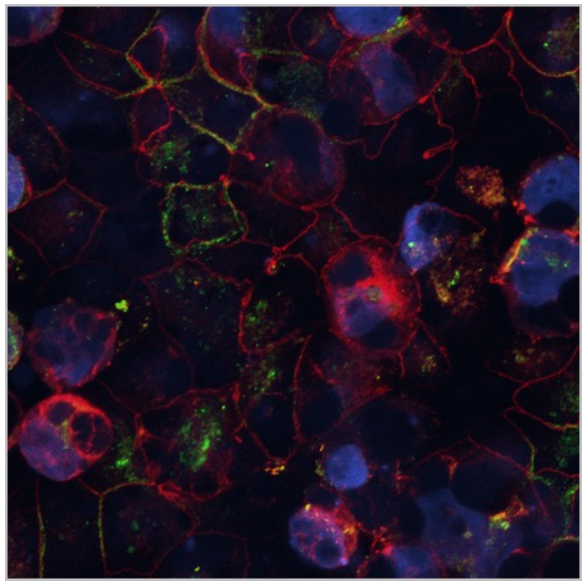
| Label | DAPI_Mean | ZO1_Mean |
|---|---|---|
| if_001.tif | 91.8 | 24.3 |
| if_004.tif | 31.5 | 3.6 |

**Then in R:** ratio = (ZO1_Mean − no-primary) ÷ DAPI_Mean, averaged per well

**Control, if_001.tif**



**P80 0.5 %, if_004.tif**



Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

# The macro, chunk by chunk



ImageJ macro

```
1  input = getDirectory("Choose the image folder");
   files = getFileList(input);
   for (i = 0; i < files.length; i++) {
     open(input + files[i]);

2    run("Z Project...", "projection=[Max Intensity]");
     img = getTitle();
     run("Split Channels");

3    selectWindow("C1-" + img);              // DAPI
     getStatistics(area, dapi);              // raw DAPI mean

4    selectWindow("C2-" + img);              // ZO-1
     getStatistics(area, zo1);               // raw ZO-1 mean

5    Table.set("Label", row, title, "ZO1");
     Table.set("DAPI_Mean", row, dapi, "ZO1");
     Table.set("ZO1_Mean", row, zo1, "ZO1");
     row++;  run("Close All");
   }
   Table.save(input + "results.csv");
```

**1 Loop over the folder**

Pick the folder once; the loop opens every image in turn. Every field gets exactly the same steps and parameters: the point of a macro.

**2 Project and split**

Max intensity over all z-slices → one 2-D image per channel: C1 = DAPI, C2 = ZO-1.

**3 DAPI mean**

No selection, so the mean covers the whole field. Raw values, no Subtract Background.

**4 ZO-1 mean**

Whole field, raw values. R subtracts the no-primary value and divides by DAPI.

**5 One row per field**

Three columns per image; after the loop the table is saved as results.csv.

# Run the macro and check its output



## results.csv: one row per field

| Label | DAPI_Mean | ZO1_Mean |
|---|---|---|
| zo1_001.tif | 419.6 | 871.4 |
| zo1_002.tif | 471.2 | 743.8 |
| zo1_046.tif | 273.4 | 303.1 |
| zo1_061.tif | 504.8 | 114.9 |

## Example dataset

4 conditions × 5 wells × 3 fields + 1 no-primary well → 63 rows (data/fiji_results_zo1.csv) + 63 rows of data/image_metadata.csv.

## Run

- Plugins › Macros › Run… → pick the image folder
- Keep image_metadata.csv complete: one row per file, written while imaging

## Check before a batch run

1. **DAPI not saturated**
   Range indicator on C1: the ratio needs real DAPI values
2. **ZO-1 not saturated**
   Brightest control junctions below the maximum value
3. **No-primary imaged the same way**
   Same laser, gain and pinhole: it is the background
4. **Metadata complete**
   One row per image in image_metadata.csv, written while imaging

# ZO-1 analysis in R

Theory, data structure, algorithm, implementation

02

# Fields are not independent: nested data






4 conditions × 5 wells × 3 fields = 60 fields, but n = 5 wells per condition

## Pseudoreplication

Counting 60 fields as n inflates the evidence: fields of one well share seeding, P80 dose, staining and imaging.

## Option 1: aggregate

Average fields → wells; compare the 5 well means per condition (ANOVA + Dunnett).

## Option 2: model the nesting

Linear mixed model: condition as fixed effect, random intercept for each well.

## Show both levels

SuperPlot: all fields in the background, well means on top (Lord et al., J Cell Biol 2020).

# Join the metadata to the results



### results.csv (Fiji)

| Label | DAPI | ZO1 |
|---|---|---|
| zo1_001.tif | 419.6 | 871.4 |
| zo1_002.tif | 471.2 | 743.8 |
| zo1_031.tif | 442.8 | 871.9 |

### image_metadata.csv (you)

| file | plate_well | condition | field |
|---|---|---|---|
| zo1_001.tif | B2 | ctrl | 1 |
| zo1_002.tif | B2 | ctrl | 2 |
| zo1_031.tif | D2 | P80-0.1 | 1 |



join by file name = Label

| file | plate_well | condition | DAPI | ZO1 | ratio |
|---|---|---|---|---|---|
| zo1_001.tif | B2 | ctrl | 419.6 | 871.4 | 1.81 |
| zo1_002.tif | B2 | ctrl | 471.2 | 743.8 | 1.34 |
| zo1_031.tif | D2 | P80-0.1 | 442.8 | 871.9 | 1.71 |

R

```r
fields <- inner_join(metadata, results,
                     by = join_by(file == Label)) %>%
  mutate(ratio = (ZO1_Mean - bg) / DAPI_Mean)   # bg: no-primary
```

### Well means: this is n

| condition | plate well | ZO-1/DAPI |
|---|---|---|
| ctrl | B2 | 1.55 |
| ctrl | B3 | 1.77 |
| ctrl | B4 | 1.98 |
| … | … | … |
| P80-0.5 | E2 | 0.61 |
| P80-0.5 | E3 | 0.95 |
| P80-0.5 | E4 | 0.65 |

### Two tables, two levels

- fields: 60 rows → mixed model
- well_means: 20 rows → ANOVA + Dunnett, SuperPlot

# Import, subtract, summarise, test, plot



**1**
**Import**
results + metadata, join by file

**2**
**Ratio**
(ZO-1 − no-primary) ÷ DAPI

**3**
**Summarise**
fields → well means

**4**
**Test**
well means or mixed model

**5**
**Plot**
SuperPlot: fields + wells

## Test what was replicated

- ANOVA + Dunnett on the 5 well means per condition (residual df = 16)
- Mixed model on all fields: random intercept per well; Kenward–Roger df = 16

## Pseudoreplication: avoid

- ANOVA + Dunnett on 60 fields: df = 56
- The p-value then reflects the number of fields you imaged, not the number of wells

# Analysis using R (1/3): from two tables to well means



```R
1 results  <- read_csv("data/fiji_results_zo1.csv") %>%
    select(-any_of(c(" ", "...1")))      # Fiji row numbers
  metadata <- read_csv("data/image_metadata.csv")
  fields <- inner_join(metadata, results,
                       by = join_by(file == Label))

2 bg <- fields %>% filter(condition == "noprimary") %>%
    pull(ZO1_Mean) %>% mean()
  fields <- fields %>% filter(condition != "noprimary") %>%
    mutate(ratio     = (ZO1_Mean - bg) / DAPI_Mean,
           condition = factor(condition, levels = conds),
           well_id   = plate_well)

3 well_means <- fields %>%
    group_by(condition, well, well_id) %>%
    summarise(ratio    = mean(ratio),
              n_fields = n(), .groups = "drop")
```

**1 Read both tables, join them by file name**

read_csv() reads Fiji's results.csv and your metadata table. inner_join() matches every Label to its file row and adds plate well, condition, well and field to each measurement.

**2 Background, then the ZO-1/DAPI ratio**

bg = mean ZO-1 of the no-primary fields: what secondary antibody and autofluorescence give without ZO-1. For every other field: ratio = (ZO-1 − bg) / DAPI, ZO-1 per nuclear DNA. factor() puts the control first, the reference for Dunnett; well_id (B2 … E6) names each well uniquely.

**3 Fields → wells: one value per replicate**

group_by() + summarise() average the 3 fields of each well: 60 field rows become 20 well rows, each with its mean ratio. n_fields checks that every well has its 3 fields. These 5 well means per condition are the n.

# Analysis using R (2/3): three tests, one right n



```R
# WRONG: 60 fields counted as n, residual df = 56
fit_fields <- aov(ratio ~ condition, data = fields)
emmeans(fit_fields, trt.vs.ctrl ~ condition)$contrasts

fit_wells <- aov(ratio ~ condition, data = well_means)
emmeans(fit_wells, trt.vs.ctrl ~ condition)$contrasts %>%
  summary(infer = TRUE)                    # df = 16

fit <- lmer(ratio ~ condition + (1 | well_id), data = fields)
anova(fit, ddf = "Kenward-Roger")
emmeans(fit, trt.vs.ctrl ~ condition,
        lmer.df = "kenward-roger")$contrasts
```

**4** **The classic mistake: fields as n**

aov() fits a one-way ANOVA; emmeans(trt.vs.ctrl) gives the Dunnett contrasts, each P80 concentration vs control. On fields the model believes it has 56 df, so the SE is too small: P80 0.1 %: p = 0.027.

**5** **Right: Dunnett on the well means**

Same model, one row per well: 20 wells − 4 groups = 16 df. summary(infer = TRUE) adds the 95 % CI of each difference. P80 0.1 %: −0.28, p = 0.11; P80 0.5 %: p < 0.001.

**6** **Right: mixed model on all fields**

lmer() keeps the 60 fields; (1 | well_id) gives every well its own intercept, so fields of one well count together. Kenward–Roger sets the df from the wells (16): the same p-values as chunk 5, and the well-to-well scatter is estimated.

# Analysis using R (3/3): the SuperPlot



```r
p <- ggplot(fields, aes(condition, ratio)) +
  geom_jitter(aes(colour = well), width = 0.15,
              alpha = 0.45, size = 1.8) +

  stat_summary(data = well_means, fun = mean, geom = "crossbar",
               width = 0.5, colour = "#333333") +

  geom_point(data = well_means, aes(fill = well, shape = well),
             size = 4.5, colour = "#333333",
             position = position_dodge(width = 0.35)) +

  scale_colour_manual(values = well_cols) +
  scale_fill_manual(values = well_cols) +
  scale_shape_manual(values = c(21, 22, 24, 23, 25)) +
  scale_x_discrete(labels = labels) +
  labs(x = NULL, y = "ZO-1 / DAPI ratio",
       colour = "Well", fill = "Well", shape = "Well",
       caption = "Simulated data; dots = fields, symbols = wells")
ggsave("figures/05_if_superplot.png", p,
       width = 6.6, height = 4.3, dpi = 200, bg = "white")
```

### 7 Fields in the background

ggplot() puts condition on x and the ratio on y. geom_jitter() draws all 60 fields, spread sideways and coloured by their well; alpha = 0.45 keeps them faint.

### 8 Bar = mean of the wells

stat_summary() on well_means, not on fields: the bar is the mean of the 5 well means, the value the test compares.

### 9 Well means on top

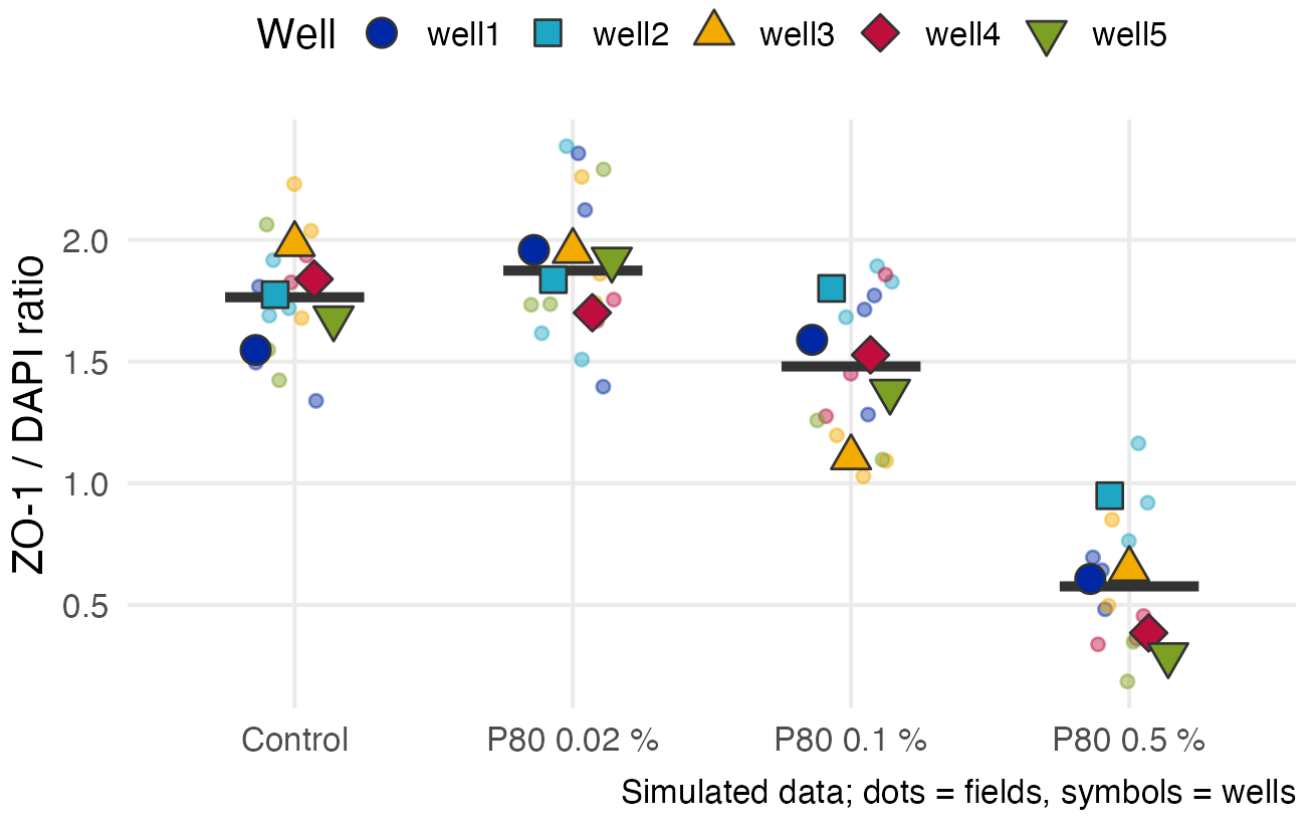
One large symbol per well, in the colour of its fields; position_dodge() sets the 5 wells side by side. These are the n.

### 10 Colours, shapes, labels, save

Each well gets one colour and one shape, so the plot also reads in grey. ggsave() writes figures/05_if_superplot.png.

# Result: the SuperPlot






## Read it in two layers

Small dots: fields; large symbols: the 5 well means per condition: the n of the test.

## ZO-1/DAPI (Dunnett, wells)

Control 1.76; 0.02 % 1.87 ($p = 0.72$); 0.1 % 1.48 ($p = 0.11$); 0.5 % 0.58 ($p < 0.001$).

## Not significant ≠ no effect

At 0.1 % the ratio is 16 % lower, 95 % CI includes 0: five wells cannot settle it.

## Next to TEER and flux

At 0.1 % TEER falls and flux rises clearly; ZO-1 per cell changes less.

# Same fields, two analyses, opposite conclusions



| ZO-1/DAPI: P80 0.1 % vs control (Dunnett) | n | Difference | df | p |
|---|---|---|---|---|
| ANOVA on all fields (pseudoreplication) | 60 fields | −0.28 | 56 | 0.027 |
| ANOVA on well means | 20 wells | −0.28 | 16 | 0.11 |
| Mixed model: random intercept per well | 60 fields in 20 wells | −0.28 ± 0.13 | 16 | 0.11 |

**0.027**

what 60 “independent” fields claim

**0.11**

what 5 wells per condition support

**Lesson**

Same estimate, opposite conclusion. Counting fields turns a well-to-well scatter into a “significant” effect.

# Before you report ZO-1 data: a checklist

1. Identical acquisition settings for every well, including the no-primary control
2. Z-stack projection stated; no saturated pixels in DAPI or ZO-1
3. Fields chosen at random on DAPI, before looking at ZO-1
4. Macro saved with the analysis; metadata table complete
5. Raw intensities, no auto-contrast; no-primary background subtracted
6. n = wells of one experiment (say so); fields averaged or modelled
7. Well means or mixed model, test named in the legend
8. SuperPlot; representative images shown with the same display range

# Run everything yourself

```
BME364-G1-share/
├── README.md          start here
├── R/                 mac_linux/, windows/ (.Rmd)
├── data/              example data (.csv)
├── example_images/    4 real IF fields (.tif)
├── fiji/              measure_zo1.ijm
├── figures/           plots, written by R
├── results/           tables, written by R
├── manuals/           lab manuals (.docx)
└── slides/            both decks (.pptx)
```

R

```r
# Fiji: Plugins > Macros > Run...
#   fiji/measure_zo1.ijm  -> results.csv

# R: R/mac_linux/ or R/windows/
# once: Source 00_install_packages.R
# open 05_if_analysis.Rmd in RStudio
#   and click Knit
```

## Your own data from Davos

- R packages: rmarkdown, tidyverse, rstatix, lme4, lmerTest, emmeans, drc
- Replace the files in data/ with your own exports and keep the column names; the .Rmd files then reproduce every step of this workshop.